

Sarvesh Prasanth Venkatesan

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EDUCATION

University of Southern California

M.S. in Electrical and Computer Engineering (Machine Learning)

Los Angeles, CA

2025 – 2027

Vellore Institute of Technology

B.Tech. in Electronics and Communication Engineering

Vellore, India

2019 – 2023

EXPERIENCE

Dynamic Robotics and Control Laboratory | University of Southern California

Graduate Researcher | Advisor: Prof. Quan Nguyen

Dec 2025 – Present

Los Angeles, CA

Visual Humanoid Loco-Manipulation

- Developing an **RL residual policy** conditioned on a **Theia visual backbone** for **whole-body loco-manipulation** across diverse objects.
- Training a **vision-language** conditioned **diffusion motion generator** for phaseless, task-directed reference motion synthesis.
- Distilled a proprioceptive expert teacher into the vision-conditioned student via **on-policy distillation with PPO objective**; deployed with ONNX runtime on the Unitree G1, reducing inference latency by ~40% on Jetson Orin.

Demonstration-Guided Dexterous Manipulation

- Designing a **demonstration-guided RL** framework for **dexterous manipulation** with a SEA module, formulating task objectives and reward shaping from VR teleoperated demonstrations.

Stochastic Robotics Lab | Indian Institute of Science

Project Associate | Advisor: Prof. Shishir Kolathaya

Jul 2024 – Aug 2025

Bengaluru, India

- Designed a **recurrent-transformer generalist** quadruped policy leveraging offline imitation learning to fuse diverse locomotion skills, achieving zero-shot **skill transfer** across 5+ unseen quadruped morphologies. [Webpage]
- Implemented **PPO-based RL locomotion** policies for the 70 kg Stoch5 quadruped; reduced **sim-to-real** gap via novel Ground Reaction Force and adaptive Raibert-inspired rewards, improving push recovery by **75–98%** on 30° low-traction inclines. [Videos]
- Implemented visual RL policies for locomotion on challenging terrains. [Code]

Mobile Robotics Lab | Indian Institute of Science

Project Associate | Advisor: Prof. Debashish Ghose

Sep 2023 – Jun 2024

Bengaluru, India

- Developed a centralized **UAV-UGV multi-agent** system for large-scale outdoor inspection, with the UAV executing coverage path planning and the UGV performing autonomous ground-level verification. [Video]
- Built onboard autonomy stacks on ROS2 using cuVSLAM for VIO, LiDAR-based obstacle avoidance, and multi-agent mission coordination in GPS-denied unstructured environments.
- Collaborated on **LocalizeSORT**, a multi-object tracking framework with world-coordinate association, achieving +15% HOTA and +30% prediction accuracy. [Paper]

Centre for Non-Destructive Evaluation | Indian Institute of Technology Madras

Intern | Advisor: Prof. Prabhu Rajagopal

Sep 2022 – Jul 2023

Chennai, India

- Developed a full software stack for a shopping-assistance robot integrating Cartographer SLAM and ROS2 Nav2 with LiDAR for autonomous retail navigation. [Video]
- Implemented MoveIt2 with OMPL for motion planning and YOLO with RGB-D depth for 3D object pose estimation, enabling end-to-end autonomous grocery pick-and-place.

PUBLICATIONS

- Narayanan PP, **Sarvesh Prasanth Venkatesan**, Srinivas Kantha Reddy, Shishir Kolathaya. GRoQ-LoCO: Generalist and Robot-agnostic Quadruped Locomotion Control using Offline Datasets. *arXiv*.
- Srihari Vemuru, Kumar Ankit, **Sarvesh Prasanth Venkatesan**, Debashish Ghose, Shishir Kolathaya. Yield Prediction and Counting Using World Coordinates. In *ASME IMECE-India 2026 Proceedings*. ASME.

PROJECTS

Manipulation VLA (Github)

- Trained a **vision-language-action** (VLA) model using a frozen LLaVA backbone with a **DiT**-based action head for multi-object manipulation in simulation.
- Benchmarked diffusion policy against flow matching on LIBERO, with flow matching achieving a higher task success rate and 2.5x inference speedup across diverse manipulation tasks.

Humanoid Locomotion (Github)

- Developed a scandot-based visual RL humanoid locomotion policy for terrain-adaptive walking. [Video]
- Trained a humanoid locomotion policy fusing egocentric depth and proprioception via **cross-modal transformer attention**, replacing privileged height scans.

Quadruped Motion Planning (Github)

- Implemented RRT* and optimization-based planners for quadruped navigation in MuJoCo, with collision checking and cost functions accounting for quadruped foothold constraints. [Video]
- Learned locomotion skills including stair climbing, slope traversal, and dynamic walking using PPO with terrain curriculum, integrated with global motion plans for hierarchical end-to-end autonomous navigation.

SKILLS

Programming: Python, C++.

ML: PyTorch, JAX, Reinforcement Learning, Imitation Learning, Diffusion, Flow Matching, VLA, sim-to-real, ONNX.

Robotics: ROS2, MoveIt, Isaac Sim, Isaac Lab, MuJoCo, mjlab, OpenCV, PX4.

Tools: Docker, Git, AWS, WandB, Fusion360, LaTeX, Ubuntu Linux.